

Arithmetic :-

- 1) Ratio (4) Average (7) Time, speed, distance (TSD)
2) Percentage (5) Mixture & Allegation (8) SI & CI (Interest)
3) Profit & Loss (6) Time & Work

* Ratio :-

1) Red: Blue = $(4:5) \times 3 = 12:15$
Green: Blue = $(2:3) \times 5 = 10:15$
 $\therefore$ Red: blue: green = $12:15:10$

2) Minimum/maximum value?

Ratio: $12:15:10 = 37n$

Minimum value = 37

Largest three digit no: $37n \leq 999$

Smallest three digit no: $37n \geq 100$

3) $\boxed{1 \text{ Large} = 5 \text{ Medium} = 7 \text{ small}} \Rightarrow 35 \text{ (Always Assume)}$

Then $1L = 35Rs, 1M = 7Rs, 1S = 5Rs$

4) If $A = \frac{15n}{7} \Rightarrow n$ should be a multiple of 7

& if $B = \frac{12n}{24} = \frac{n}{2} \Rightarrow n$ should be a multiple of 2

$\therefore n = \text{LCM}(7, 2) = 14$

5) $C \propto \omega^2$

$\Rightarrow \boxed{C = k\omega^2}$

where $C = \text{Cost}$
 $\omega = \text{weight}$
 $k = \text{Constant}$

(6) $E_v = kn$

$E_T = E_v + E_F$

$\Rightarrow \boxed{E_T = kn + E_F}$

$n = \text{no. of student}$
 $E_F = \text{fixed cost}$
 $E_v = \text{variable cost}$
 $E_T = \text{Total cost}$

(7) $a+b+c=10$ (Given)

$(a^2+b^2+c^2)_{\max}?$

$\Rightarrow a=8, b=1, c=1$

(values as far as possible)

$(a^2+b^2+c^2)_{\min}?$

$\Rightarrow a=3, b=3, c=4$

(values as close as possible)

Close-min
far-max

* Concept: —

$\frac{a, b, c}{S.P} \rightarrow (a^2 + b^2 + c^2) \max \uparrow$
 Loss min $\downarrow \rightarrow$ far $\rightarrow (a^2 + b^2 + c^2) \max \uparrow$
 Loss max $\uparrow \rightarrow$ close $\rightarrow (a^2 + b^2 + c^2) \min \downarrow$

* Concept: —

$$\frac{A}{B} = \frac{11}{14} \Rightarrow \left(\frac{11}{14}\right) \times 3 = \frac{33}{42}$$

$$\Rightarrow \frac{A+n}{B+n} = \frac{47}{56} \Rightarrow \left(\frac{11}{14}\right) \times 3 = \frac{33}{42}$$

Make the difference of numerator & denominator same

$$\therefore n = 14 \checkmark$$

Ex: — $\frac{A}{B} = \frac{5}{7} \Rightarrow \left(\frac{5}{7}\right) \times 7 = \frac{35}{49}$

$$\frac{A+n}{B+n} = \frac{32}{39} \Rightarrow \left(\frac{32}{39}\right) \times 2 = \frac{64}{78}$$

$$\therefore \left(\frac{35}{49} : \frac{64}{78}\right) 29 \therefore n = 29 \checkmark$$

* If four factors given in Question, make table: —

	1 Bedroom	2 Bedroom
Rented	40	240
NonRented	160	50

You can answer anything by this table.

$$\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{a+c+e}{b+d+f}$$

$$\frac{a}{b} = \frac{c}{d} \text{ (given)}$$

$$\Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d} \text{ (Componendo & Dividendo)}$$

$$a \propto \frac{1}{b} \Rightarrow a = \frac{k}{b}$$

$$\Rightarrow \boxed{ab = k}$$

$$a \times b = k$$

100 100 10000 (Assume)
 $\downarrow$ $\downarrow$ $\downarrow$
 200 50 10000 (A/Q)

Percentage : —

$$\begin{aligned}
 1 &= 100\% \\
 1/2 &= 50\% \\
 1/3 &= 33.33\% \\
 1/4 &= 25\% \\
 1/5 &= 20\% \\
 1/6 &= 16.66\% \\
 1/7 &= 14.28\% \\
 1/8 &= 12.5\% \\
 1/9 &= 11.1\% \\
 1/10 &= 10\%
 \end{aligned}$$

Subtraction rule: —

$$\begin{aligned}
 11\% - 7\frac{2}{3}\% \\
 &= (10+1)\% - (7+\frac{2}{3}\%) \\
 &= 3\% + \frac{1}{3}\% \\
 &= 3\frac{1}{3}\%
 \end{aligned}$$

Multiplication rule: —

$$\begin{aligned}
 (16\frac{1}{6}\%) \times 5 \\
 &= (16+\frac{1}{6}) \times 5 = (80+\frac{5}{6}) = 80\frac{5}{6}\%
 \end{aligned}$$

* If $A = 120, B = 150$

A is what % of B $\Rightarrow \frac{A}{B} \times 100\% = \frac{120}{150} \times 100\% = 80\%$

* A is what % greater than B $\Rightarrow \frac{A-B}{B} \times 100\%$

* $N = 80 \xrightarrow{20\% \text{ incr}} 80 \times \frac{120}{100} = 96$

$N = 120 \xrightarrow{30\% \text{ decr}} 120 \times \frac{70}{100} = 84$

* Always assume 100, x, 100 in questions ($100\% \Rightarrow \text{VVIP}$)

* per capita $\times$ population = Total Production

$$\begin{array}{c}
 100x \\
 \downarrow \text{A/Q } 27\% \\
 127x
 \end{array}$$

$$\begin{array}{c}
 100y \\
 \downarrow \\
 14000y \\
 127x
 \end{array}$$

$$\begin{array}{c}
 10000y \\
 \downarrow \text{A/Q } 40\% \\
 14000y
 \end{array}$$

$\therefore \text{increase in population} = \frac{14000y}{127x} - \frac{1000y}{1} = \frac{13000y}{127}$

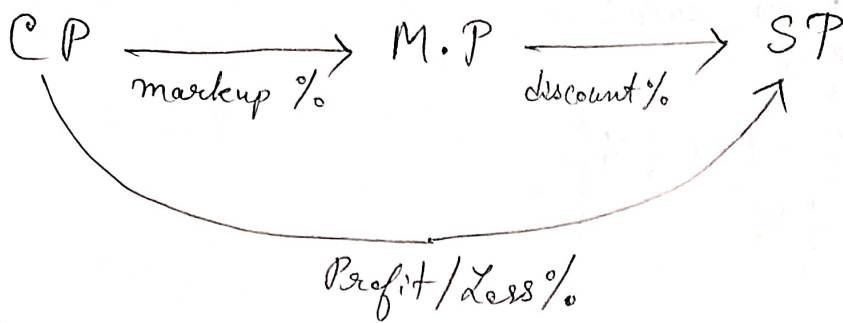
* Four factors $\rightarrow$ Always make Table

* Always assume in such a way that it makes calculations simpler

Ex: - Q48) Total egg = $9x + 2$ (let) $\rightarrow 9x \leftarrow \begin{array}{l} \text{fridge (5x)} \\ \text{Remaining (4x)} \end{array}$

① % by which A exceeds B = $\frac{A - B}{B} \times 100\%$

Profit & Loss : ———



$$*) \text{ Loss} = \text{C.P.} - \text{S.P.} \quad \left| \begin{array}{l} \text{Profit/Loss} \\ \text{Loss \%} \end{array} \right. = \frac{\text{Profit/Loss}}{\text{C.P.}} \times 100\%$$

$$*) \text{ Profit} = \text{S.P.} - \text{C.P.}$$

$$*) \text{ S.P.} = \text{M.P.} - \text{discount}$$

$$*) \text{ SP}(10A) = \text{CP}(19A) = \frac{1}{3} D(70A) = 70 \text{ Rs (Let)}$$

$$*) \text{ Labelling} \rightarrow \text{M.P.} \quad (\text{Always Assume})$$

$\begin{array}{c} A \\ 100n \\ \downarrow \\ 120n \end{array}$	$\begin{array}{c} B \\ 100y \text{ (C.P.)} \\ \downarrow \\ 90y \text{ (S.P.)} \end{array}$	$\left \begin{array}{c} A \\ 300 \\ \downarrow \\ 360 \end{array} \right.$	$\begin{array}{c} B \\ 400 \text{ (C.P.)} \\ \downarrow \\ 360 \text{ (S.P.)} \end{array}$
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A/a — $120n = 90y$
 $\Rightarrow 4n = 3y$ if $n=3$
 $\Rightarrow y = \frac{4n}{3}$ $y=4$ (substitute them $\uparrow$)

$\begin{array}{c} Q \\ 11 \rightarrow 10 \text{ Rs} \\ 1 \rightarrow \frac{10}{11} \text{ Rs} \\ \text{(C.P.)} \end{array}$	$\left \begin{array}{c} R_s \\ 10 \rightarrow 11 \text{ Rs} \\ 1 \rightarrow \frac{11}{10} \text{ Rs} \\ \text{(S.P.)} \end{array} \right.$	$\begin{array}{l} \text{Profit} = \frac{11}{10} - \frac{10}{11} = \frac{21}{110} \\ \text{Profit\%} = \frac{21/110}{10/11} \times 100\% \end{array}$
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$\begin{array}{c} Q \\ 11 \rightarrow 10 \text{ Rs} \end{array}$	$\times 10$	$\left \begin{array}{c} R_s \\ 10 \rightarrow 11 \text{ Rs} \end{array} \right.$	$\times 11$
$\Rightarrow 100 \rightarrow 100 \text{ Rs}$		$\Rightarrow 110 \rightarrow 121 \text{ Rs}$	
(Same Quantity) $\therefore \text{Profit} = 121 - 100 = 21$			

$$\star) \text{ Revenue} = \text{Market Share} \times \overset{\text{Unit}}{\text{S.P}}$$

$$\text{Profit} = \underset{(\%)}{\text{Profitability}} \times \text{Revenue}$$

*)	Syrup	Juice
<u>Q</u>	100 kg	120 kg
<u>P</u>	₹ 800/kg	₹ 1000/kg

$$C.P = ₹ 88000 + ₹ 120000 = 20,8000 \text{ Rs}$$

*) when summation given, assume $\begin{cases} x \text{ Rs.} \\ (50,000 - x) \text{ Rs} \end{cases}$

*) Successive increment :-

$$100 \xrightarrow{a\%} 120 \xrightarrow{b\%} 132$$

$$\text{overall increment \%} = \left(a + b + \frac{ab}{100} \right) \%$$

*) Successive decrement :-

$$100 \xrightarrow{-a\%} 90 \xrightarrow{-b\%} 81$$

$$\text{overall decrement \%} = \left\{ (-a) + (-b) + \frac{ab}{100} \right\} \%$$

*) Case : — (if $a = +ve \text{ incr}$, $b = -ve \text{ decr}$)

$$100 \xrightarrow{+a\%} 120 \xrightarrow{-b\%} 108$$

$$\begin{aligned} \text{overall \%} &= a + (-b) + \frac{a(-b)}{100} \\ \text{incr/decr \%} &= \left(a - b - \frac{ab}{100} \right) \% \end{aligned}$$

*) if two candidates stand in election, one wins the elections by 400 votes

Let Total votes = 1000

A	B	[Assumption technique]
500 + 200	500 - 200	
(winning candidate)	(losing candidate)	

Asson the election by 400 votes. (Assumption rule ↑)

Averages :-

* $\text{Avg} = \frac{\text{TS (Total Sum)}}{n \text{ (no. of objects)}}$

* $\boxed{\text{TS} = \text{Avg} \times n} \text{ (VVIP)}$

Find TS of every numerical in Average chapter
as TS = attack point

*)

	n	Avg	TS
Boys	20	x	$20x$
Girls	30	$(x+5)$	$30(x+5)$
Total	50	$(x+2)$	$50x+150$

[follow this format]

*) Movement from Past $\rightarrow$ future (Age Numericals)

Ex:-

10 year ago	$\xrightarrow{n}$	$\xrightarrow{\text{Avg}}$	$\xrightarrow{\text{TS}}$
	6	$\frac{\text{Avg}}{n}$	$\frac{TS}{6n}$
$\downarrow$	$\downarrow$		$\downarrow + (10 \times 6)$
Current	$6-1+1 = 6$	Avg	$6n+60-F+B$
		$\Rightarrow \frac{6n+60-F+B}{6} = n(A/a)$	

*) $\text{min value} \leq \text{Avg} \leq \text{Maximum value}$ (important)

*) Avg of 30 integers is 5 $\Rightarrow \text{TS} = 30 \times 5 = 150$

But Avg of 20 integers is ≤ 5 & Avg of 10 integers is > 5

30

20 ≤ 5

TS = $150 - 60 = 90$

Avg = $90/20 = 4.5$

10	> 5	TS = 60
Let 6	minimize	

if one side has to be maximized,
then the other side must be
minimized. (VVIP)

* $A_1 \leq A_2 \leq A_3 \dots \leq A_9 \Rightarrow \text{mean} = 42 \text{ (given)}$

Then, $A_1 \text{ max} \rightarrow 42$
 $A_9 \text{ min} \rightarrow 42$

$A_2 \leq A_3 \leq A_4 \dots \leq A_{10} \Rightarrow \text{mean} = 47 \text{ (given)}$

Then, $A_2 \text{ max} \rightarrow 47$
 $A_{10} \text{ min} \rightarrow 47$

Max Avg = $\frac{A_1 + A_2 + \dots + A_{10}}{10} = \frac{42 + (9 \times 47)}{10}$

Min Avg = $\frac{(9 \times 42) + A_{10}}{10} = \frac{(9 \times 42) + 47}{10}$

for max Avg, substitute $A_1 \text{ max}$

for min Avg, substitute $A_{10} \text{ min}$

* $B \leq G \leq 2B$

$G_{\text{max}} = 2B$

$G_{\text{min}} = B$

* Weight of E is 12 Kg more than D

$\Rightarrow E - D = 12$

(Remember this concept $\uparrow$)

214) A says to B "I am twice as old as you were when I was as old as you were!"

	Present	Past
A	$2x$	y
B	y	x

(Tricky Questions)
 (SNAP, NMAT)

Time & work : —

*) Efficiency $\times$ No. of days = Tw (Total work)

*) $\boxed{\text{Efficiency} = \frac{\text{Tw}}{\text{days}}}$ (VVIP) [Attack point: — Efficiency]

*) Always assume Tw in order to find efficiency [days = given]

*) Efficiency = $\frac{1}{\text{No. of days}}$ [Tw = Constant]

Efficiency and Time are inversely proportional (opposite)

Ex $\frac{A_T}{B_T} = \frac{4}{5}$ then $\rightarrow \frac{A_{\text{eff}}}{B_{\text{eff}}} = \frac{5}{4}$

*) Always assume Total work (Tw) as the LCM of days.

*) Money will always be distributed in ratio of Tw [of unequal days]

$A_{Tw} = 60 \text{ unit}$ $A : B$ $\therefore A = \frac{1}{3} \times 1500 = \text{₹} 500$
 $B_{Tw} = 120 \text{ unit} \Rightarrow 1 : 2$ $\therefore B = \frac{2}{3} \times 1500 = \text{₹} 1000$
Total money = ₹1500

No. of days $\neq$ same $\rightarrow$ Work ratio $\rightarrow$ money distribute

No. of days = same $\rightarrow$ efficiency ratio $\rightarrow$ money distribute

*) $\frac{Tw}{V} = \text{Constant}$ Always [Tw = Total work, V = Volume of work]

$\Rightarrow \frac{Tw_1}{V_1} = \frac{Tw_2}{V_2}$ $\left\{ \begin{array}{l} \% (A+B) \text{ can complete a work in } n \text{ days} \\ \% A \text{ alone can " " " in } n+m \text{ days} \\ \% B \text{ alone can " " " in } n+n \text{ days} \end{array} \right.$

*) Then, (A+B) can complete the work in $\boxed{n = \sqrt{mn}}$ days ✓

*) Concept $\rightarrow$ efficiency can be added
Time/days can't be added [Always Convert Time $\rightarrow$ efficiency]

* Concept :-

$$T_w = 12 \text{ unit}$$

inlet pipe ($I_{\text{efficiency}}$) = 3 u/hr (Always +ve)

outlet pipe ($O_{\text{efficiency}}$) = -4 u/hr (Always -ve)

Ex: - $I_{\text{eff}} \times 30 = T_w$

$$(5I_{\text{eff}} + 4O_{\text{eff}}) \times 10 = T_w$$

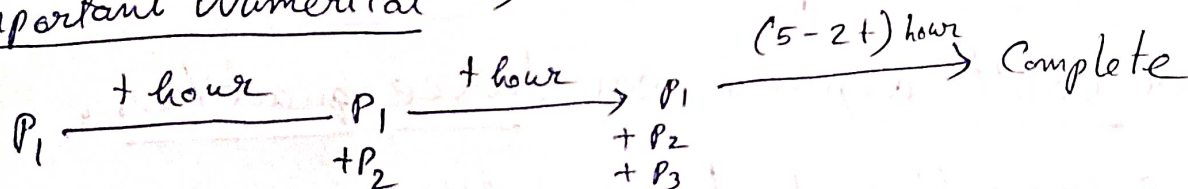
(Here, outlet efficiency is taken positive at first)

Equate: - $I_{\text{eff}} \times 30 = (5I_{\text{eff}} + 4O_{\text{eff}}) \times 10$

$$\Rightarrow I_{\text{eff}} = -2O_{\text{eff}} \leftarrow \text{eventually } O_{\text{eff}} \text{ is -ve}$$

* $Q_{\text{share}} = \frac{Q_{\text{work}}}{T_w} \times \text{Total money}$

Important Numerical $\rightarrow$ Total time = 5 hours, $T_w = 60$ (given)



$$(P_1 \times t) + (P_1 + P_2) \times t + (P_1 + P_2 + P_3) \times (5 - 2t) = T_w = 60$$

* $(A+B) \rightarrow n \text{ days}$
 $A \text{ alone} \rightarrow n+m \text{ days}$
 $B \text{ alone} \rightarrow n+n \text{ days}$

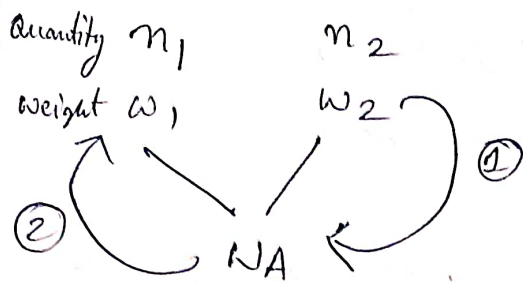
$$\therefore n = \sqrt{mn} \text{ days}$$

<u>Assumption Rule</u>		
$A_{\text{ef}} = 5n$		$A_{\text{ef}} = 5$
$B_{\text{ef}} = 4n$	O R	$B_{\text{ef}} = 4$
$T_w = 20$		$T_w = 20n$

Then, frame the equation

* Mixture & Allegation

Derivation:



$$\frac{n_1}{n_2} = \frac{w_2 - w_A}{w_A - w_1} \text{ (VVIP)}$$

	Class 1	Class 2
Quantity	n_1	n_2
Weight	w_1	w_2
	$n_1 w_1$	$n_2 w_2$
	w_A	
	$w_A = \frac{n_1 w_1 + n_2 w_2}{n_1 + n_2}$	

Ex: -

	M_1	M_2
Quantity	40L	n L
Percentage	10%	100%

$\swarrow \quad \searrow$
 $M_3 (20\%)$

$$\therefore \frac{40}{n} = \frac{100 - 20}{20 - 10}$$

$$\Rightarrow w_A (n_1 + n_2) = n_1 w_1 + n_2 w_2$$

$$\Rightarrow n_1 w_A - w_1 n_1 = n_2 w_2 - n_2 w_A$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{w_2 - w_A}{w_A - w_1}$$

Weight, percentage, Cost/Kg, Avg fraction $\Rightarrow$ Applicable (ratio)

* Allegation can't applied on three quantities. Allegation only applies to two quantities.

Ex: - A(3) B(1) (Important variety)

$m: w$ $m: w$
 $2: n$ $4: 1$

$$w = \frac{n}{2+n}$$

$$w = \frac{1}{5}$$

$m: w$
 $1: 1$
 $w = \frac{1}{2}$

$$\therefore \frac{A}{B} = \frac{\frac{1}{5} - \frac{1}{2}}{\frac{1}{2} - \frac{n}{2+n}}$$

$\Rightarrow$ Solving we get,
 $n = 3$

Multiplication factor

$$1) MF = \frac{\text{final}}{\text{initial}} ; MF = \frac{\text{Remaining}}{\text{original}} \quad \left| \quad \frac{SP}{CP} = \frac{\text{Real weight}}{\text{false weight}} \right.$$

$$2) MF = \frac{SP}{CP} \quad \left| \quad SP = CP \times MF_1 \times MF_2 \times \dots \quad (VVIP) \right.$$

Profit: $MF > 1$

Loss: $MF < 1$

Ex:- uses 800 gm instead of 1000

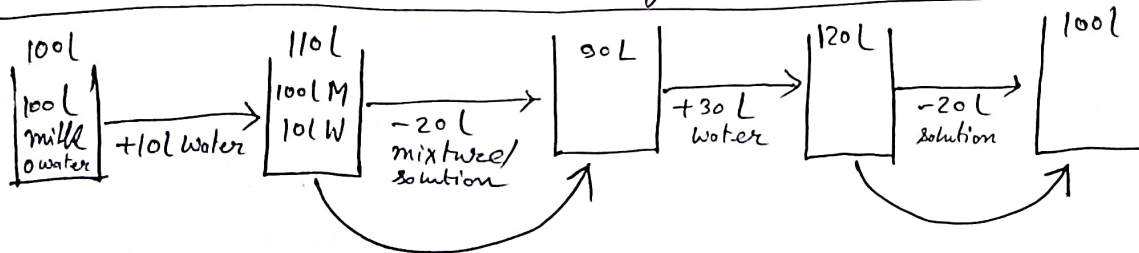
$$SP = C.P \times \frac{1000}{800}$$

uses $\rightarrow$ denominator
instead $\rightarrow$ numerator

instead $1000 \text{ gm} \rightarrow$ uses 800 gm
 $MF = \frac{1000}{800}$

~~Buy: cheats by 100 gm instead of a Kg~~
 ~~$MF = \frac{1100}{1000}$~~
 ~~$1100 \text{ gm} \rightarrow 1000 \text{ gm}$~~

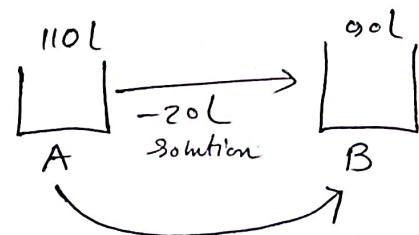
Dishonest seller $\rightarrow$ Always use MF or traditional method



Water is changing (add/sub) $\rightarrow$ we can find Milk Final (vice versa)

$$M_{\text{Final}} = M_{\text{initial}} \times MF_1 \times MF_2$$

Places where we are extracting solution
(jekhane jekhane solution ber korchhi)
(jaha par solution nikal raha hun)



$\rightarrow$ Make an Arrow $\Rightarrow MF = \frac{B}{A} = \frac{90}{110}$

* Use this above concept where equal quantity is taken out & added

* Concept:- A dishonest dealer cheats his supplier & buyer by 100 gms instead of a Kg:-

Buy: $1000 \text{ gm} \rightarrow 1100 \text{ gm}$ | Sell $\rightarrow 1000 \text{ gm} \rightarrow 900 \text{ gm}$

VVI

1) $1L = 5M = 7S = 35$
(Always Assume)

2) Population $\times$ per capita = production
 $100u \quad y \quad 100uy$
 $\downarrow \quad \downarrow 27\% \quad \downarrow 90\%$
 $? \quad 1.27y \quad 90uy$

$? \rightarrow \frac{90uy}{1.27y} = \text{Ans}$

3) % by which A exceeds B = $\frac{A - B}{B} \times 100\%$

4) $0.4W = 0.5B$

$\Rightarrow \frac{B}{W} = \frac{4}{5}$ | Assume $\longrightarrow$
 $B = 4$
 $W = 5$

$$5) \quad 120u = 00y \quad \left| \quad \begin{array}{l} \text{Assume } u = 3 \\ y = 4 \end{array} \right.$$

$$\Rightarrow \frac{u}{y} = \frac{3}{4}$$

$$6) \quad (2M + 9B) \times 8 = (4M + 2B) \times 6$$

solve — — —

$$\Rightarrow \frac{M}{B} = \frac{5}{2}$$

Assume — — —

$$B = 2u/d$$

$$M = 5u/d$$

$$*) \quad \frac{Tw_1}{v_1} = \frac{Tw_2}{v_2}$$

$$v_2 \text{ or } v_1 = \frac{100 \text{ cm} \cdot \text{t}}{221 \text{ cm} / 3600 \text{ pages etc}}$$

$$\textcircled{2/7} / \textcircled{5/7} / \text{etc.}$$

$$*) \quad \frac{\text{Time}}{2u} = \frac{\text{eff}}{2u}$$

* Allegation ~~3 quantity~~ (Not applicable)

* $C.P \times C.P \mid C.P \times 1.2 = S.P$

$C.P \mid \Rightarrow C.P = S.P \times \frac{10}{12}$

* 20% impurity $\mid$ Profit

$$S.P = C.P \times \frac{120}{100}$$

* Cheating in Buying $\rightarrow 100 \rightarrow 125 \left(\frac{125}{100} \right)$

Cheating in selling $\rightarrow 100 \rightarrow 75 \left(\frac{100}{75} \right)$

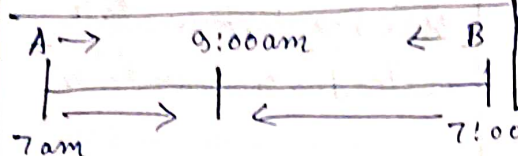
$$S.P = C.P \times \frac{125}{100} \times \frac{100}{75}$$

* Time, Speed, Distance (TSD) (VVIP)

$$1) D = S \times t \quad || \quad S = \frac{D}{t} \quad || \quad t = \frac{D}{S}$$

$$2) \text{ If } D \text{ is constant: } \left| \text{ If } S \text{ is constant: } \left| \text{ If } T \text{ is constant: } \right. \right.$$

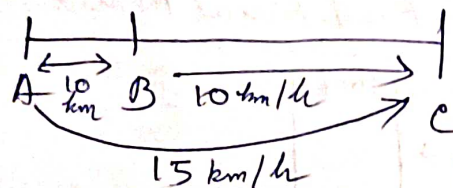
$$\checkmark \frac{S_1}{S_2} = \frac{T_2}{T_1} \left(S \propto \frac{1}{T} \right) \quad \frac{D_1}{D_2} = \frac{T_1}{T_2} \left(D \propto T \right) \quad \frac{D_1}{D_2} = \frac{S_1}{S_2} \left(D \propto S \right)$$


 (meets at same time after starting at same time)

(*) Time constant (Meeting) : —

$$\Rightarrow \frac{D_1}{D_2} = \frac{S_1}{S_2} \quad (t \text{ is constant})$$

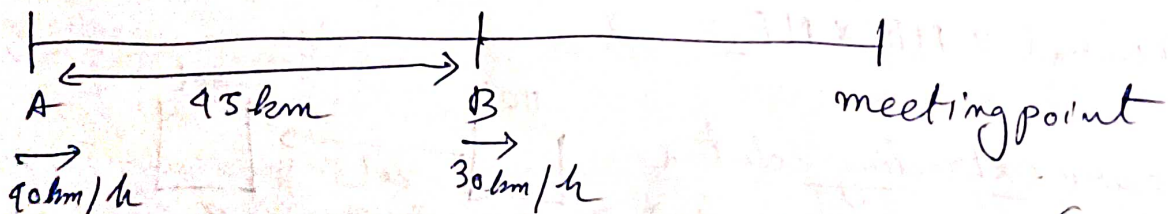
* Catching : — (As Time is same / constant)


 $\Rightarrow \frac{S_A}{S_B} = \frac{D_A}{D_B} = \frac{15}{10} = \frac{3}{2}$

* When A will catch B?

A/q, $15d - 10d = 10$ | $3d - 2d = 10$
 $\Rightarrow 5d = 10$ | $\Rightarrow d = 10$
 $\Rightarrow d = 2$

$T_A = 3d / 15 = 30 / 15 = 2 \text{ hours (Answer)}$



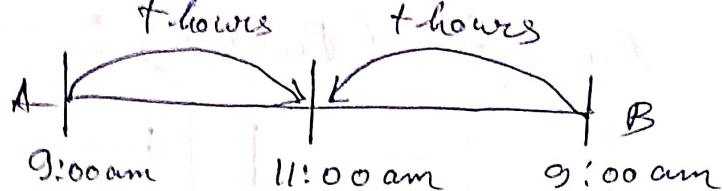
	Meeting Time	Distance
30 km/h	$\rightarrow t \text{ hour} \rightarrow$	$30t$
40 km/h	$\rightarrow t \text{ hour} \rightarrow$	$40t$

(Catching Concept)

A/q — $40t - 30t = 45$
 $\Rightarrow 10t = 45$
 $\Rightarrow t = 4.5 \text{ hour } \checkmark$

Meeting Concept: —

(time constant)



Relative Speed: —

$\xrightarrow{V_2} \quad \xleftarrow{V_1}$
(Opposite direction)

$$\Rightarrow R.S = V_1 + V_2$$

$\xrightarrow{V_2} \quad \xrightarrow{V_1}$
(Same direction)

$$\Rightarrow R.S = V_2 - V_1$$

* time = $\frac{\text{Gap Distance}}{R.S.}$ (VVIP) $\Rightarrow$

* Extra distance = speed $\times$ Extra time

* Escalator: —

effective speed = $(1 + e)$ or $(1 - e)$ step/sec

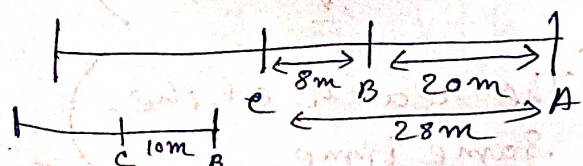
[$\because e$ = speed of escalator, 1 = speed of man]

Total step = $\frac{\text{Effective (s)}}{\text{speed}} \times \text{time}$ ✓
(D)

Time = $\frac{\text{Total steps (D)}}{\text{effective speed}}$

Time = $\frac{\text{Steps}}{\text{Human Speed}}$ ✓

* Linear Race: — To find length of race track ?

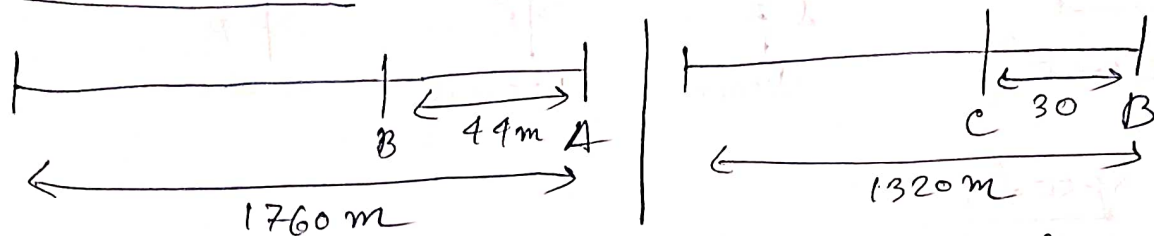


A beats B by 20 m
A beats C by 28 m [given]
B beats C by 10 m

Gap needs to travel
B-C (20) $\rightarrow$ B (20 m more)
B-C (10) $\rightarrow$ B (10 m)
B-C (10) $\rightarrow$ B (100 m)

$\therefore$ length of race = 100 m

Linear Race :- (Part 2) : —



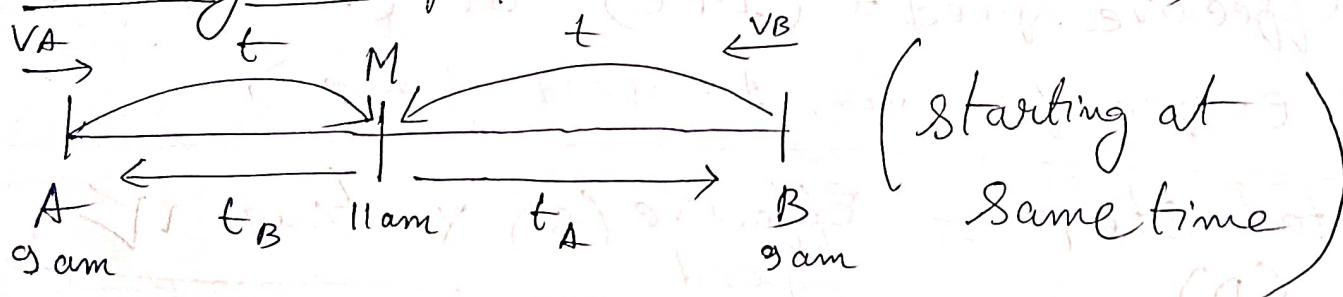
A beats B by 44m & B can beat C by 30m
in a 1760m race | in a 1320m race
(given) | (given)

Find the distance will A beat C in a 880m race?

$$\text{Ans :- } \frac{V_A}{V_B} \times \frac{V_B}{V_C} = \frac{1760}{1716} \times \frac{1320}{1290}$$

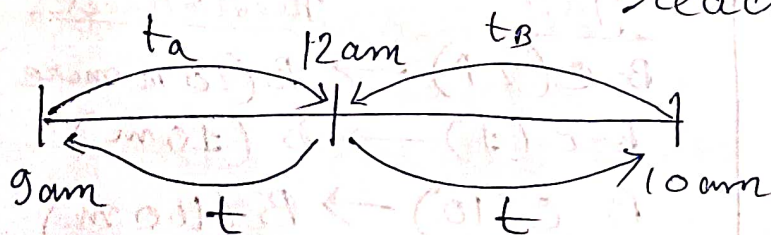
$$\Rightarrow \frac{V_A}{V_C} = \frac{880}{838.5} \therefore \text{A will beat C by } 41.5\text{m}$$

Meeting Concept : — (ratio of speed)



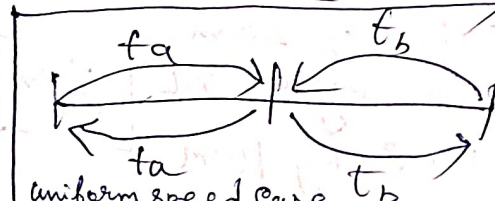
$$1) \frac{V_A}{V_B} = \sqrt{\frac{t_B}{t_A}} \quad \text{[reaching at different time]}$$

$$2) t = \sqrt{T_A T_B} \quad \text{(time taken to reach the end points)}$$

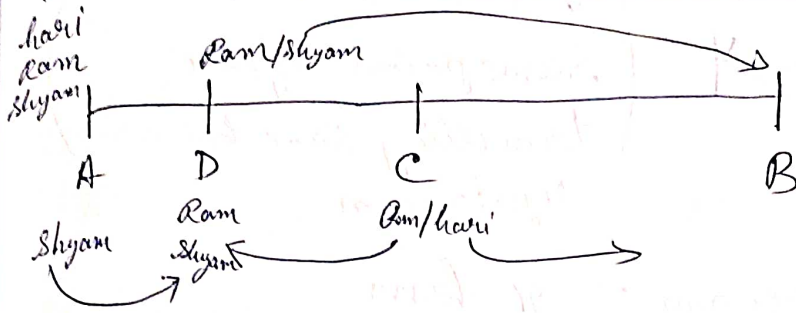


(starting at different time)

(The reach at the same time)



Meeting of Three persons : —



$$\begin{aligned} V_S &= 1 \text{ km/h} \\ V_R &= 5 \text{ km/h} \\ V_H &= 1 \text{ km/h} \end{aligned}$$

Case 1 : —

Ram : A to C to D } Same time
Shyam : A to D

$$\frac{V_S}{V_R} = \frac{D_S}{D_R} = \frac{1}{5}$$

Case 2 : —

Ram : C to D to B } same time
Hari : C to B

$$\frac{V_R}{V_H} = \frac{D_R}{D_H} = \frac{5}{1}$$

$$\therefore AD = 1, DC = 2, CB = 1 \text{ (Distance)}$$

Boat & Stream : —

$$\begin{aligned} \text{Boat (speed)} &\rightarrow V \\ \text{Stream (speed)} &\rightarrow u \end{aligned} \quad \left| \quad \begin{aligned} \text{Downstream} &= (V + u) \\ \text{Upstream} &= (V - u) \end{aligned} \right.$$

fixed object floating = u (speed of stream) [Ex: - Wood]

Always frame equations : — ($T_D + T_U = \text{Total Time}$)

$$\Rightarrow \frac{D}{V_D} + \frac{D}{V_U} = 14 \quad \left(\begin{aligned} &\text{Play with facts of } D (48) \\ &= (1, 2, 3, 4, 6, 8, 12, 16, 24, 48) \end{aligned} \right)$$

$$\Rightarrow \frac{48}{V_D} + \frac{48}{V_U} = 14$$

$$V_D = 8 \Rightarrow V + u = 8$$

$$V_U = 6 \Rightarrow V - u = 6$$

$$\therefore V = 7 \text{ km/h}$$

$$u = 1 \text{ km/h}$$

$$\frac{44}{V_D} + \frac{36}{V_U} = 16$$

$$\frac{55}{V_D} + \frac{40}{V_U} = 13$$

Common factor

$$(44, 55) = 11 (V_D)$$

H.C.F

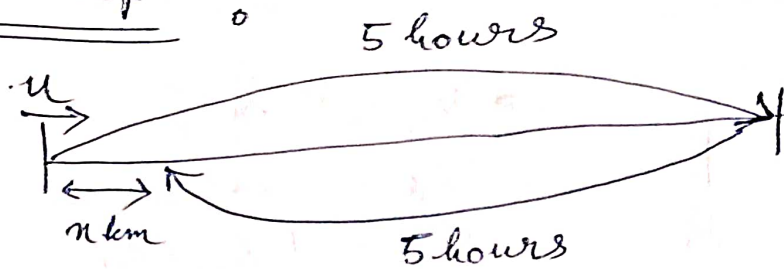
$$V + u = 11$$

$$V - u = 3$$

$$V = 8 \text{ km/h}$$

$$u = 3 \text{ km/h}$$

Concept: —

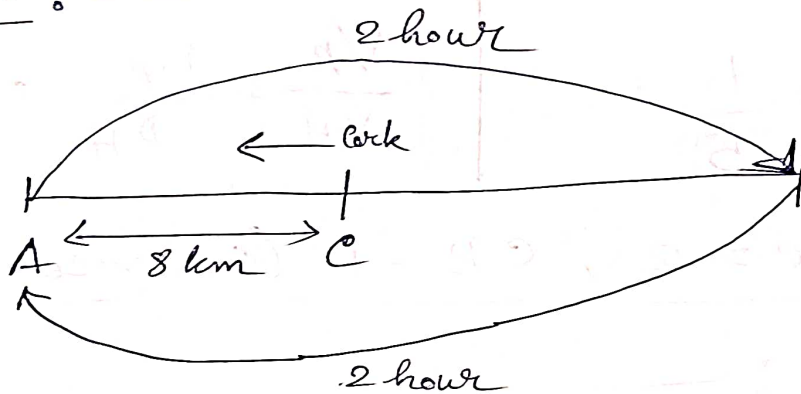


(will not return to same point after travelling same hours upstream)

$$D_{\text{Downstream}} - D_{\text{Upstream}} = n \text{ km}$$

$$\Rightarrow (v+u) \times 5 - (v-u) \times 5 = n \text{ km} \checkmark$$

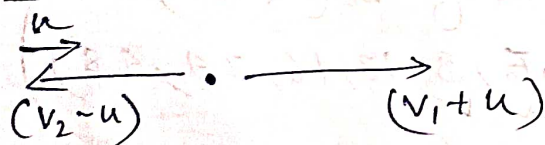
Concept: —



$$\text{Cork C to A (time)} = \frac{8}{2+2} = \frac{8}{4} = 2 \text{ hours}$$

$$*) T_D + T_U = \text{Total} \& T_U - T_D = \text{Difference Time}$$

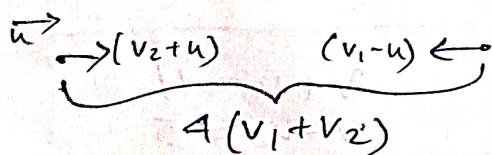
$$\text{Ex: } \frac{D}{v+u} + \frac{D}{v-u} = 14 \quad \left| \quad \frac{D}{v-u} - \frac{D}{v+u} = 6 \right.$$



Two rowers row for 4 hours at different direction

$$R.S = v_1 + u + v_2 - u = v_1 + v_2 \checkmark \quad \text{Time } T = 4 \text{ hours (given)} \quad (T_1)$$

$$\text{gap} = S \times t = (v_1 + v_2) \times 4$$



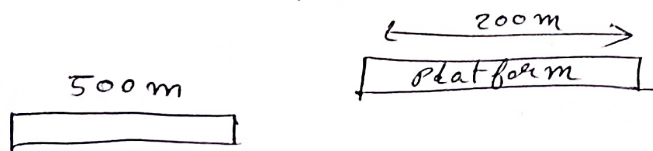
They start moving towards each other

$$\text{Time} = \frac{4(v_1+v_2) \text{ gap}}{v_1+v_2 \text{ RS}} = 4 \text{ hours} \checkmark \quad (T_2)$$

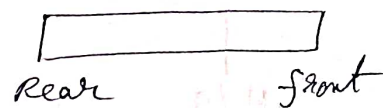
$$\therefore T_1 = T_2 \text{ (fact)}$$

$$R.S = v_1 + v_2 \checkmark$$

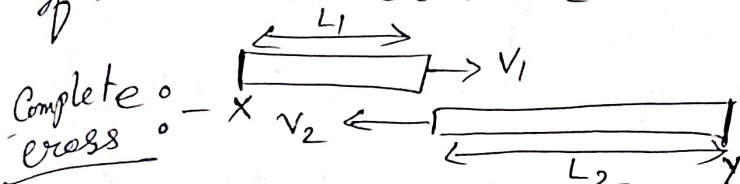
Train Concept : —



$$\text{Total length} = 200 + 500 = 700 \text{ m}$$

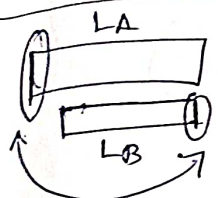


If two train cross one another



Complete cross

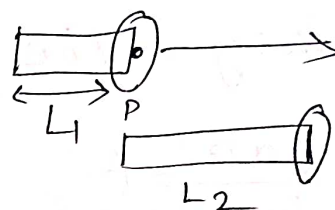
$$\frac{L_1 + L_2}{V_1 + V_2} = 14 \text{ (time)}$$



$$\frac{L_A}{R.S} = \text{time}$$

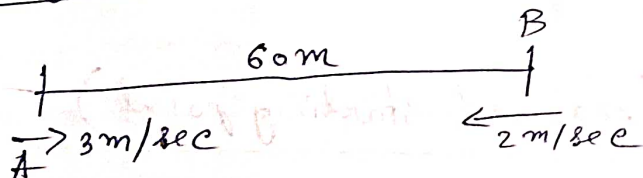
Passenger Case : —

Passenger always sit in first seat.



$$\frac{L_2 (\text{gap})}{R.S} = \text{time}$$

To & fro : — (speed ratio < 2) & $= 2$)



Ex: — Total Distance in 5th meeting

$$= D + (2D \times 4) = 9D = 9 \times 60 = 540 \text{ m}$$

$$1^{\text{st}} \text{ meet} = D$$

$$2^{\text{nd}} \text{ meet} = 2D$$

$$3^{\text{rd}} \text{ meet} = 2D$$

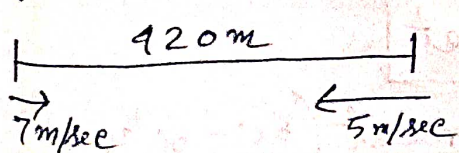
$$4, 5, 6, \dots n^{\text{th}} \text{ meet} = 2D$$

Distance: — $A = \frac{3}{5} \times 540$, $B = \frac{2}{5} \times 540$

$$\frac{V_A}{V_B} = \frac{3}{2} \quad \checkmark$$

(speed ratio > 2) $\Rightarrow$ Geometrical Approach (Later)

If A Covered 3600 m distance, find no. of meeting?



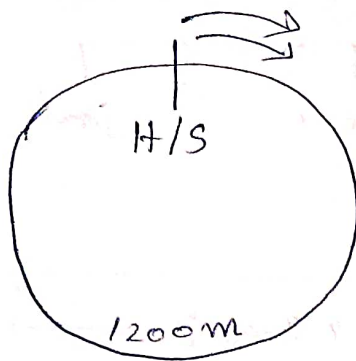
$$TD = D + (n-1) \times 2D \text{ (distance eq)}$$

$$A_D = \frac{7}{12} \{ D + (n-1) \times 2D \}$$

$$B_D = \frac{5}{12} \{ D + (n-1) \times 2D \}$$

(speed ratio ≤ 2)
Here, < 2

Circular Race :-



$$H_s = 150 \text{ m/sec}$$

$$S_s = 80 \text{ m/sec}$$

a) when they will meet together at starting point?

$$\text{Ans} - \text{LCM}(t_1, t_2)$$

$$= \text{LCM}\left(\frac{1200}{150}, \frac{1200}{80}\right) = 120 \text{ sec}$$

b) when they will be together (1st meet) for first time?

$$\text{Ans} :- \frac{\text{gap}}{\text{R.S}} = t$$

$$\Rightarrow \frac{1200}{150 - 80} = t \Rightarrow t = \frac{120}{7} \text{ sec (1st meet)}$$

$$2^{\text{nd}} \text{ meet} \rightarrow \frac{120}{7} \times 2 \text{ sec}$$

$$3^{\text{rd}} \text{ meet} \rightarrow \frac{120}{7} \times 3 \text{ sec}$$

...

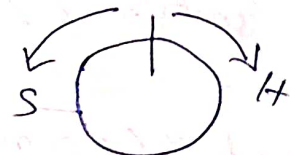
$$7^{\text{th}} \text{ meet} \rightarrow \frac{120}{7} \times 7 = 120 \text{ sec (reached starting point)}$$

c) No. of distinct meeting point?

$$\text{Ans} :- \frac{V_H}{V_S} = \frac{150}{80} = \frac{15}{8} \Rightarrow 7 \left[\text{Same direction} \right]$$

7 meeting points (distinct)

$$\frac{V_H}{V_S} = \frac{150}{80} = \frac{15}{8} \Rightarrow 23 \left[\text{opposite direction} \right]$$



Distance travelled by H & S for 1st meet

$$D_S = S \times t = 80 \times \frac{120}{7} \text{ m}$$

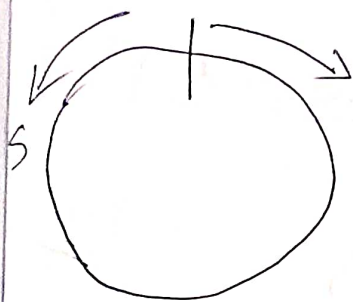
$$D_H = S \times t = 150 \times \frac{120}{7} \text{ m}$$

$$D_H - D_S = 1 \text{ round } \checkmark (\Rightarrow)$$

for n^{th} meet (Ex:- 60th meet)

$$D_S = 80 \times \frac{120}{7} \times 60 \text{ m}$$

$$D_H = 150 \times \frac{120}{7} \times 60 \text{ m}$$



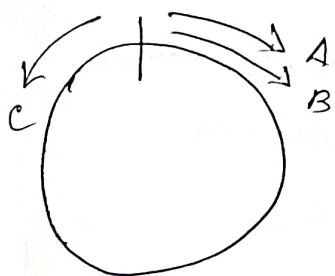
H 1) $D_H + D_S = 1 \text{ round}$ (opposite $\leftarrow \rightarrow$)

2) distinct meeting point = $\frac{15}{8} \times 23$

3) Together 1st meet (Time) = $\frac{1200}{150+80} = \frac{1200}{230} \text{ sec}$

4) 100th meet will coincide with which no. of meeting?

Ans: — $100 \bmod 23$



$V_A = 150$

$V_B = 70$

$V_C = 110$

$\frac{V_A}{V_B} = \frac{150}{70} = \frac{15}{7} \Rightarrow 8 \text{ meeting point}$

$\frac{V_A}{V_C} = \frac{150}{110} = \frac{15}{11} \Rightarrow 26 \text{ meeting point}$

$\frac{V_B}{V_C} = \frac{70}{110} = \frac{7}{11} \Rightarrow 18 \text{ meeting point}$

How many times all 3 will meet? $\Rightarrow \text{HCF}(8, 18, 26) = 2 \text{ distinct meet points}$

$V_A : V_B : V_C = 15 : 7 : 11$

Distance for 100th meet

2 meet $\rightarrow A \xrightarrow{\text{travel}} 15 \text{ rounds} \Rightarrow \frac{15}{2} \text{ Round} \times 100 \text{ m}$

2 meet $\rightarrow B \xrightarrow{\text{travel}} 7 \text{ round} \Rightarrow \frac{7}{2} \text{ Round} \times 100 \text{ m}$

2 meet $\rightarrow C \xrightarrow{\text{travel}} 11 \text{ round} \Rightarrow \frac{11}{2} \text{ Round} \times 100 \text{ m}$

Concept: —

If distinct meet point = even: Diametrically opposite possible

If distinct meet point = odd: Diametrically opposite (X)
not possible

Ex: — 4, 6 $\rightarrow$ Diametrically opposite meet points (✓)

3, 7 $\rightarrow$ Diametrically opposite (X)
not possible

$$*) \frac{V_A}{V_B} = \frac{11}{7} \quad \& \quad (\text{meet points moving in same direction}) \quad *)$$

4 meet $\rightarrow$ A $\xrightarrow{\text{travel}}$ 11 rounds (Remember)

4 meet $\rightarrow$ B $\xrightarrow{\text{travel}}$ 7 rounds

Additional Concepts: —

River: —

*) when Total time given (T is given): —

$$T_D + T_U = T \quad \left| \begin{array}{l} \text{generally, we will get two equations.} \\ \text{solve them to get the values} \\ \text{of } V \& U. \end{array} \right.$$

$$\Rightarrow \frac{D}{V+U} + \frac{D}{V-U} = T$$

*) when Difference Time (t is given): —

$$T_U - T_D = t$$

$$\Rightarrow \frac{D}{V-U} - \frac{D}{V+U} = t$$

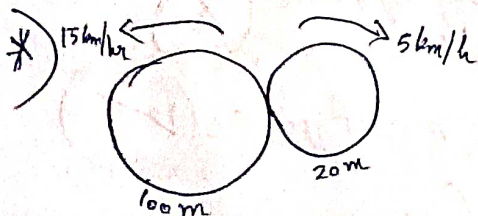
$$*) \quad \begin{array}{|c|c|c|} \hline 60 & 60 & 60 \\ \hline 10 \text{ km/h} & 20 \text{ km/h} & 30 \text{ km/h} \\ \hline \end{array} \quad \text{Bimal}$$

Assumption when D is divided into 3 equal parts

$$\begin{array}{|c|c|c|} \hline t & t & t \\ \hline 10 \text{ km/h} & 20 \text{ km/h} & 30 \text{ km/h} \\ \hline \end{array} \quad \text{Amal}$$

Assumption when The path is divided into 3 equal time frames.

*) Distance = Circumference $\times$ no. of revolutions
travelled by wheel (Perimeter)



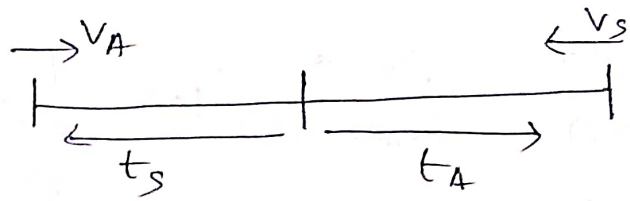
$$T_{\text{Rahim}} = \frac{2\pi}{2500}, \quad T_{\text{Ram}} = \frac{2\pi}{150}$$

$$\frac{T_{\text{Ram}}}{T_{\text{Rahim}}} = \frac{5}{3} \quad \left| \begin{array}{l} 5, 10, 15, \dots \\ 3, 6, 9, 12, 15, \dots \end{array} \right.$$

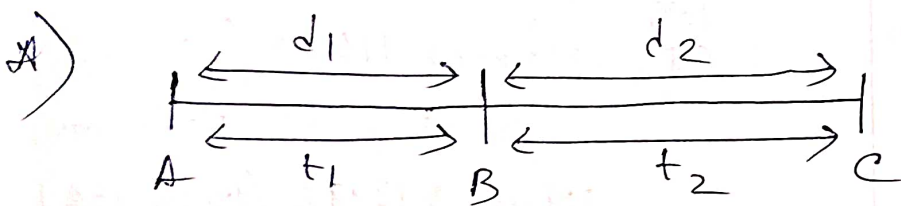
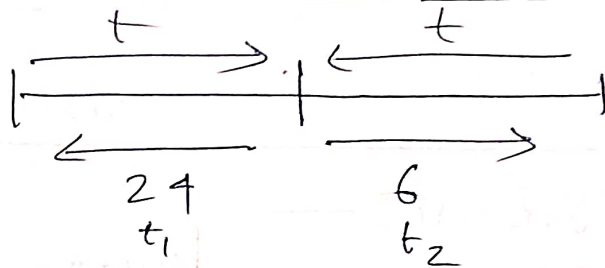
Time when they meet = LCM ($T_{\text{Ram}}, T_{\text{Rahim}}$) = 15

* Meeting concept Revise: — (VVIP)

$$\frac{V_{\text{Arrive}}}{V_{\text{Start}}} = \sqrt{\frac{t_{\text{Start}}}{t_{\text{Arrive}}}}$$



$$t = \sqrt{t_1 t_2}$$



$$\text{Avg speed} = \frac{d_1 + d_2}{t_1 + t_2} \quad \text{or} \quad \frac{d_1 + d_2}{\frac{d_1}{v_1} + \frac{d_2}{v_2}}$$

if $d_1 = d_2 \Rightarrow \text{Avg speed} = \frac{2v_1 v_2}{v_1 + v_2}$ (same as H.P series)

Simple & Compound Interest

$$SI = \frac{PRT}{100}$$

Rs 100 (Principal) $\xrightarrow{10\%}$ 1st year $\rightarrow$ 10 Rs (10% of base amount)
 2nd year $\rightarrow$ 10 Rs (10% of base amount)
 3rd year $\rightarrow$ 10 Rs (10% of base amount)
 3 year $\rightarrow$ 30 Rs (30% of base amount)
 Total Interest

$$A = P + I$$

For compound interest: —

$$A = P \left(1 + \frac{r}{100}\right)^n$$

where n is Time
 $r = \text{rate}$
 $P = \text{Principal}$

$$A - P = I$$

Rs 1000 $\xrightarrow{10\%}$ 1st year $\rightarrow$ 100 Rs (10% on base)
 CI 2nd year $\rightarrow$ 110 Rs (11% on base)
 3rd year $\rightarrow$ 121 Rs (12.1% on base)
 4th year $\rightarrow$ 133 Rs (13.3% on base)

Compound Interest
 $\rightarrow$ interest on interest

when CI is 10 %

1st year $\rightarrow$ 10%
 2nd year $\Rightarrow 10\% + 10 \times \frac{10}{100} = 11\%$
 3rd year $\Rightarrow 11\% + 11 \times \frac{10}{100} = 12.1\%$
 4th year $\Rightarrow 12.1\% + \frac{12.1 \times 10}{100} = 13.3\%$

1st year = 10%
 2nd year = 11%
 3rd year = 12.1%
 4th year = 13.3%
 $\Rightarrow$ 21%
 $\Rightarrow$ 33.1%
 $\Rightarrow$ 46.4%

Most Important $\uparrow$

For 1 year $\Rightarrow$ 10% CI is same as 10% SI

Sum of SI for 3 years if SI = 12% ~~= 36%~~
 $= 3 \times 12\% = 36\%$ on base (12 + 12 + 12 %)

Sum of CI for 3 years if CI = 10%
 $= 10\% + 11\% + 12.1\% = 33.1\%$ on base

Journey of $A_2 \rightarrow A_4$ in CI: — (VVIP)

$$A_2 \times \left(1 + \frac{R}{100}\right)^2 = A_4 ; A_2 \times \left(1 + \frac{R}{100}\right)^3 = A_5$$

Solving of equation in CI: —

$$8000 = P \left(1 + \frac{R}{100}\right)^2 \dots (1)$$

$$2700 = P \left(1 + \frac{R}{100}\right)^5 \dots (2)$$

Divide them to find
 Value of R

*) Payable half yearly (Semiannually) / Quarterly case :-

Principal = 10,000 Rs

Rate = 6% (Semiannually)

Time = 2 years

Semiannually - 1 year = 2 cycles

Quarterly - 1 year = 4 cycles

$$\text{New rate} = \frac{6}{\text{No of cycles in 1 year}} = \frac{6}{2} = 3\%$$

$$\text{No. of cycles} = (2 \times 2) = 4 \text{ cycles}$$

$$\begin{aligned} A &= P \left(1 + \frac{R}{100} \right)^n \\ &= 10,000 \left(1 + \frac{3}{100} \right)^4 \end{aligned}$$

*) Concept :- Extra money = Extra interest

Concept: — (for CI)

$$I_4 = I_3 + I_3 \times \frac{r}{100} \quad \text{where } I_1 = \text{Interest of 1st year}$$

$$I_3 = \text{Interest of 3rd year}$$

$$\Rightarrow I_4 - I_3 = I_3 \times \frac{r}{100} \quad \checkmark$$

When something is divided in two parts & there exists a condition of equality.

Assume $\Rightarrow$ Money

(100n & 100y)
rule

100n 100y (Let)
15% 12%
4 years 3 year
60n 36y

A/Q —

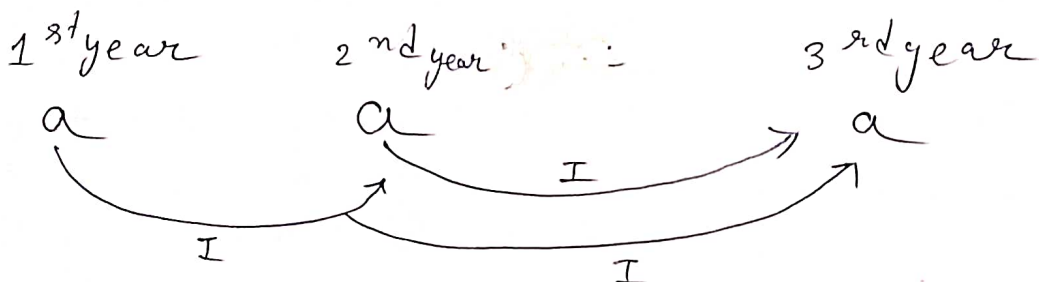
$$60n = 36y$$

$$\therefore y = 5, n = 3$$

put these & solve

Installments: — Method 1 is of house.

Method 2: — 500 Rs, 10%, 3 equal installment



VVIP

$$a \left(1 + \frac{10}{100}\right)^2 + a \left(1 + \frac{10}{100}\right)^1 + a = A \quad \checkmark$$

Method 3: — Play with rate (r%)

10% (2 installments)

$$\frac{10\%}{1} = \frac{10}{100} = \frac{1}{10}$$

Rs	Rs	Rs	Rs
10 $\rightarrow$	11	$\Rightarrow$ 110 $\rightarrow$	121
$(10)^2 \rightarrow$	121	$\Rightarrow$ +100 $\rightarrow$	121
		210 $\rightarrow$	121 Rs
		Rs 21000 $\rightarrow$	12100 Rs
			(Ans)

10% (3 installment) installment

Rs	Rs	Rs	Rs
10 $\rightarrow$	11	1210 $\rightarrow$	1331
$(10)^2 \rightarrow$	121	$\Rightarrow$ 1100 $\rightarrow$	1331
$(10)^3 \rightarrow$	1331	1600 $\rightarrow$	1331

method 2
is always
better
approach

Rs 3310 $\rightarrow$	Rs 1331
Rs 1 $\rightarrow$	Rs 1311
	3310
Rs n $\rightarrow$	Rs 1331 $\times$
	3310

Partnership

Profit $\xrightarrow{\text{gets divided in the ratio of}}$ Investment $\times$ time

$$\frac{P_A}{P_B} = \frac{2000 \text{ Rs} \times 5}{3000 \text{ Rs} \times 6} = \frac{10^5}{180} = \frac{5}{9}$$

$$P_A = \frac{5}{14} \times \text{total Profit}$$

$$P_B = \frac{9}{14} \times \text{total Profit}$$
